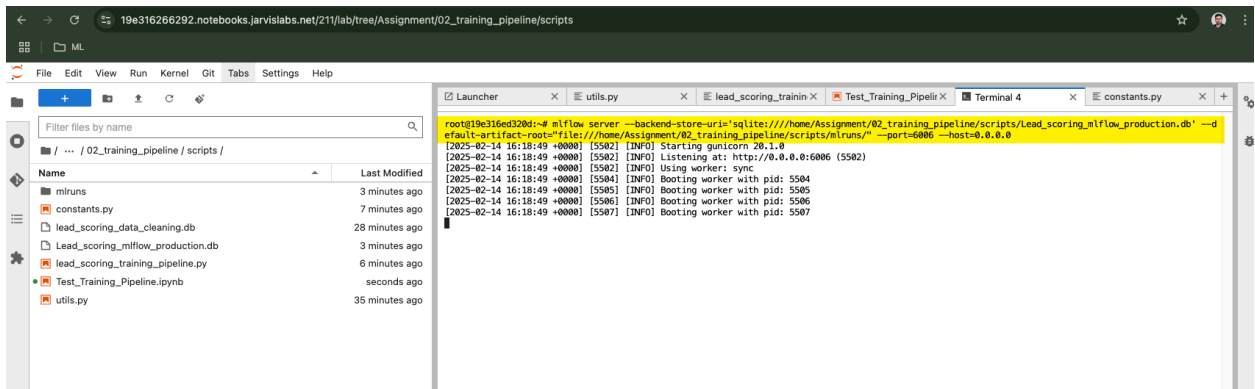


ML OPS Training Pipeline

Tejas Kapasi - ML C63
KapasiTejas@gmail.com

Running the ML Flow Server



The screenshot shows a JupyterLab interface with a file browser on the left and a terminal on the right. The file browser displays the directory structure of the ML Flow server, including files like `constants.py`, `lead_scoring_data_cleaning.db`, `lead_scoring_mlflow_production.db`, `lead_scoring_training_pipeline.py`, `Test_Training_Pipeline.ipynb`, and `utils.py`. The terminal window shows the command `mlflow server --backend-store-uri=sqlite:///home/Assignment/02_training_pipeline/scripts/Lead_scoring_mlflow_production.db` being executed, followed by logs indicating the server is listening at `http://0.0.0.0:6006` and booting workers.

```
root@19e316e320d:~# mlflow server --backend-store-uri=sqlite:///home/Assignment/02_training_pipeline/scripts/Lead_scoring_mlflow_production.db --default-artifact-root=file:///home/Assignment/02_training_pipeline/scripts/mlruns/ --port=6006 --host=0.0.0.0
[2025-02-14 16:18:49 +0000] [5502] [INFO] Starting gunicorn 20.1.0
[2025-02-14 16:18:49 +0000] [5502] [INFO] Listening at: http://0.0.0.0:6006 (5502)
[2025-02-14 16:18:49 +0000] [5502] [INFO] Using worker: sync
[2025-02-14 16:18:49 +0000] [5504] [INFO] Booting worker with pid: 5504
[2025-02-14 16:18:49 +0000] [5505] [INFO] Booting worker with pid: 5505
[2025-02-14 16:18:49 +0000] [5506] [INFO] Booting worker with pid: 5506
[2025-02-14 16:18:49 +0000] [5507] [INFO] Booting worker with pid: 5507
```

Experiments - Lead_scoring_mlflow_production

19e3162662921.notebooks.jarvislabs.net/#/experiments/2

mlflow 1.26.1 Experiments Models GitHub Docs

Experiments

Search Experiments

- Default
- Lead_Scoring_Train...
- Lead_scoring_mlflow...

Lead_scoring_mlflow_production

Share

Track machine learning training runs in experiments. [Learn more](#)

Experiment ID: 2

Description [Edit](#)

[Refresh](#) [Compare](#) [Delete](#) [Download CSV](#) [Start Time](#) [All time](#)

[Columns](#) Only show differences [Search](#) [Filter](#) [Clear](#)

Showing 1 matching run

	Start Time	Duration	Run Name	User	Source	Version	Models	Metrics	Parameters
☐	21 seconds ago	8.8s	run_LightGB	root	ipykerne...	-	LightGBM/4	False Negative: 6190, Precision: 0.739, Precision_0: 0.789	boosting_type: gbd, class_weight: None

[Load more](#)

19e3162662921.notebooks.jarvislabs.net/#/experiments/2/runs/93fe3ca12aa648519d5c5e735a91a479

mlflow 1.26.1 Experiments Models GitHub Docs

Lead_scoring_mlflow_production > run_LightGB

run_LightGB

Date: 2025-02-14 22:01:06 Source: ipykernel_launcher.py User: root

Duration: 8.8s Status: FINISHED Lifecycle Stage: active

Description [Edit](#)

Parameters (20)

Metrics (12)

Tags

Artifacts

models

- MLmodel
- conda.yaml
- model.pkl
- python_env.yaml
- requirements.txt

Full Path: file:///home/Assignment/02_training_pipeline/scripts/mlruns/2/93fe3ca12aa648519d5c5e735a91a479/artifacts/models

LightGBM, v4 Registered on 2025/02/14

MLflow Model

The code snippets below demonstrate how to make predictions using the logged model. This model is also registered to the [model registry](#).

Model schema

Input and output schema for your model. [Learn more](#)

Name	Type
No schema. See MLflow docs for how to include input and output schema with your model.	

Make Predictions

Predict on a Spark DataFrame:

```
import mlflow
logged_model = 'runs:/93fe3ca12aa648519d5c5e735a91a479/models'

# Load model as a Spark UDF. Override result_type if the model does not return double values.
loaded_model = mlflow.pyfunc.spark_udf(spark, model_uri=logged_model, result_type='double')

# Predict on a Spark DataFrame.
columns = list(df.columns)
df.withColumn('predictions', loaded_model(*columns)).collect()
```

19e3162662921.notebooks.jarvislabs.net/#/models/LightGBM

ML

mlflow 1.26.1 Experiments Models GitHub Docs

Registered Models > LightGBM

LightGBM

Created Time: 2025-02-14 21:49:18 Last Modified: 2025-02-14 22:17:18

Description Edit

Tags

Versions All Active 1 Compare

☐	Version	Registered at	Created by	Stage	Description
☐	☒ Version 4	2025-02-14 22:01:13		Production	
☐	☒ Version 3	2025-02-14 21:59:14		None	
☐	☒ Version 2	2025-02-14 21:49:42		None	
☐	☒ Version 1	2025-02-14 21:49:18		None	

<

1

>

19e3162662921.notebooks.jarvislabs.net/#/experiments/2/runs/93fe3ca12aa648519d5c5e735a91a479

ML

mlflow 1.26.1 Experiments Models GitHub Docs

Lead_scoring_mlflow_production > run_LightGB

run_LightGB

Date: 2025-02-14 22:01:06 Source: ipykernel_launcher.py User: root

Duration: 8.8s Status: FINISHED Lifecycle Stage: active

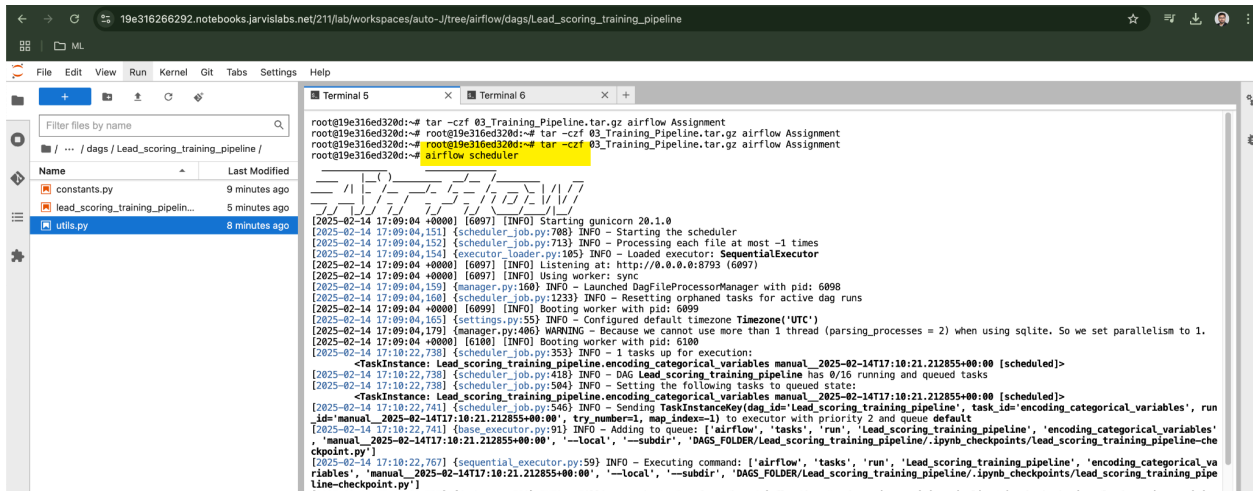
Description Edit

Parameters (20)

Metrics (12)

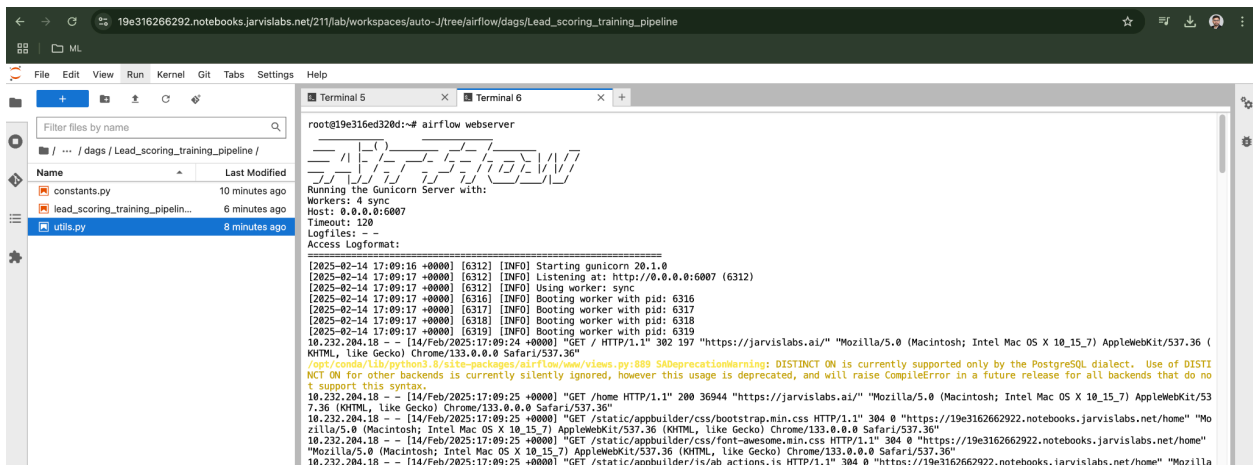
Name	Value
False Negative	6190
Precision	0.739
Precision_0	0.789
Precision_1	0.706
Recall	0.747
Recall_0	0.649
Recall_1	0.829
True Negative	23095
auc	0.747
f1_0	0.712
f1_1	0.762

Airflow Server Running for Training Pipeline



```
root@19e316ed320d:~# tar -czf 03_Training_Pipeline.tar.gz airflow Assignment
root@19e316ed320d:~# root@19e316ed320d:~# tar -czf 03_Training_Pipeline.tar.gz airflow Assignment
root@19e316ed320d:~# root@19e316ed320d:~# tar -czf 03_Training_Pipeline.tar.gz airflow Assignment
root@19e316ed320d:~# airflow scheduler

[2025-02-14 17:09:04.151] [scheduler_job.py:708] INFO - Starting the scheduler
[2025-02-14 17:09:04.152] [scheduler_job.py:713] INFO - Processing each file at most -1 times
[2025-02-14 17:09:04.154] [executor_loader.py:105] INFO - Loaded executor: SequentialExecutor
[2025-02-14 17:09:04.160] [6097] [INFO] Listening at: http://0.0.0.0:8793 (6097)
[2025-02-14 17:09:04.160] [6097] [INFO] Using worker: sync
[2025-02-14 17:09:04.150] [manager.py:160] INFO - Launched DagFileProcessorManager with pid: 6098
[2025-02-14 17:09:04.160] [scheduler_job.py:1233] INFO - Resetting orphaned tasks for active dag runs
[2025-02-14 17:09:04.160] [6099] [INFO] Booting worker with pid: 6099
[2025-02-14 17:09:04.165] [settings.py:55] INFO - Configured default timezone Timezone('UTC')
[2025-02-14 17:09:04.179] [manager.py:406] WARNING - Because we cannot use more than 1 thread (parsing_processes = 2) when using sqlite. So we set parallelism to 1.
[2025-02-14 17:09:04.180] [6100] [INFO] Booting worker with pid: 6100
[2025-02-14 17:10:22.738] [scheduler_job.py:353] INFO - 1 tasks up for execution:
-taskInstance: Lead_scoring_training_pipeline.encoding_categorical_variables manual_2025-02-14T17:10:21.212855+00:00 [scheduled]>
[2025-02-14 17:10:22.738] [scheduler_job.py:418] INFO - DAG Lead_scoring_training_pipeline has 0/16 running and queued tasks
[2025-02-14 17:10:22.738] [scheduler_job.py:504] INFO - Setting the following tasks to queued state:
-taskInstance: Lead_scoring_training_pipeline.encoding_categorical_variables manual_2025-02-14T17:10:21.212855+00:00 [scheduled]>
[2025-02-14 17:10:22.741] [scheduler_job.py:546] INFO - Sending TaskInstanceKey(dag_id='Lead_scoring_training_pipeline', task_id='encoding_categorical_variables', run_id='manual_2025-02-14T17:10:21.212855+00:00', try_number=1, map_index=-1) to executor with priority 2 and queue default
[2025-02-14 17:10:22.741] [base_executor.py:91] INFO - Adding to queue: ['airflow', 'tasks', 'run', 'Lead_scoring_training_pipeline', 'encoding_categorical_variables', 'manual_2025-02-14T17:10:21.212855+00:00', '--local', '--subdir', 'DAGS_FOLDER/Lead_scoring_training_pipeline/.ipynb_checkpoints/Lead_scoring_training_pipeline-checkpoint.py']
[2025-02-14 17:10:22.767] [sequential_executor.py:59] INFO - Executing command: ['airflow', 'tasks', 'run', 'Lead_scoring_training_pipeline', 'encoding_categorical_variables', 'manual_2025-02-14T17:10:21.212855+00:00', '--local', '--subdir', 'DAGS_FOLDER/Lead_scoring_training_pipeline/.ipynb_checkpoints/Lead_scoring_training_pipeline-checkpoint.py']
```



```
root@19e316ed320d:~# airflow webserver

Running the Unicorn Server with:
Host: 0.0.0.0:8080
Timeout: 120
Logfiles: -
Access LogFormat:

[2025-02-14 17:09:16.000] [6312] [INFO] Starting unicorn 20.1.0
[2025-02-14 17:09:17.000] [6312] [INFO] Listening at: http://0.0.0.0:8080 (6312)
[2025-02-14 17:09:17.000] [6312] [INFO] Using worker: sync
[2025-02-14 17:09:17.000] [6316] [INFO] Booting worker with pid: 6316
[2025-02-14 17:09:17.000] [6317] [INFO] Booting worker with pid: 6317
[2025-02-14 17:09:17.000] [6318] [INFO] Booting worker with pid: 6318
[2025-02-14 17:09:17.000] [6319] [INFO] Booting worker with pid: 6319
10.232.204.18 - - [14/Feb/2025:17:09:24 +0000] "GET / HTTP/1.1" 302 197 "https://jarvislabs.ai/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
/opt/conda/lib/python3.8/site-packages/airflow/www/views.py:889 SAWarning: DISTINCT ON is currently supported only by the PostgreSQL dialect. Use of DISTINCT ON for other backends is currently silently ignored, however this usage is deprecated, and will raise CompileError in a future release for all backends that do not support this syntax.
10.232.204.18 - - [14/Feb/2025:17:09:25 +0000] "GET /home HTTP/1.1" 200 36944 "https://jarvislabs.ai/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
10.232.204.18 - - [14/Feb/2025:17:09:25 +0000] "GET /static/appbuilder/css/bootstrap.min.css HTTP/1.1" 304 0 "https://19e3162662922.notebooks.jarvislabs.net/home" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
10.232.204.18 - - [14/Feb/2025:17:09:25 +0000] "GET /static/appbuilder/css/font-awesome.min.css HTTP/1.1" 304 0 "https://19e3162662922.notebooks.jarvislabs.net/home" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
10.232.204.18 - - [14/Feb/2025:17:09:25 +0000] "GET /static/appbuilder/js/ab_actions.js HTTP/1.1" 304 0 "https://19e3162662922.notebooks.jarvislabs.net/home" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
```

19e3162662922.notebooks.jarvislabs.net/home

Airflow DAGs Security Browse Admin Docs 17:11 UTC TK

Do not use SQLite as metadata DB in production – it should only be used for dev/testing. We recommend using Postgres or MySQL. [Click here](#) for more information.

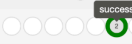
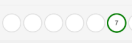
Do not use SequentialExecutor in production. [Click here](#) for more information.

DAGs

All 34 Active 2 Paused 32

Filter DAGs by tag

Search DAGs

DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions	Links
 Lead_scoring_training_pipeline	airflow		@monthly	2025-02-14, 17:10:21	2025-03-01, 00:00:00		 	...
 TK_Lead_Scoring_Data_Engineering_Pipeline	airflow		@daily	2025-02-14, 14:31:34	2025-02-14, 00:00:00		 	...

Showing 1-2 of 2 DAGs

19e3162662922.notebooks.jarvislabs.net/dags/Lead_scoring_training_pipeline/grid

Airflow DAGs Security Browse Admin Docs 17:10 UTC TK

DAG: Lead_scoring_training_pipeline

Training pipeline for Lead Scoring System

Schedule: @monthly Next Run: 2025-03-01, 00:00:00

Grid Graph Calendar Task Duration Task Tries Landing Times Details <> Code Audit Log

14/02/2025, 05:10:39 PM 25 All Run Types All Run States Clear Filters

Auto-refresh

Duration: 00:00:23

encoding_categorical_variables
training_model

DAG Lead_scoring_training_pipeline

DAG Details

DAG Runs Summary

Total Runs Displayed	1
Total running	1
First Run Start	2025-02-14, 17:10:22 UTC
Last Run Start	2025-02-14, 17:10:22 UTC
Max Run Duration	00:00:23
Mean Run Duration	00:00:23
Min Run Duration	00:00:23

DAG Summary

Total Tasks	2
PythonOperators	2

19e3162662922.notebooks.jarvislabs.net/dags/Lead_scoring_training_pipeline/graph

Airflow DAGs Security Browse Admin Docs 17:11 UTC TK

DAG: Lead_scoring_training_pipeline Training pipeline for Lead Scoring System success Schedule: @monthly Next Run: 2025-03-01, 00:00:00

Grid Graph Calendar Task Duration Task Tries Landing Times Gantt Details <> Code Audit Log

2025-02-14T17:10:22Z Runs 25 Run manual__2025-02-14T17:10:21.212855+00:00 Layout Left > Right Update Find Task...

PythonOperator

deferred failed queued running scheduled skipped success up_for_reschedule up_for_retry upstream_failed no_status

Auto-refresh

```
graph LR; encoding_categorical_variables --> training_model;
```